

Pattern A

Immunization Write Path — V2 FHIRconnect

Full FHIR Bundle → COMPOSITION-level openFHIR mapping → openEHR → EHRbase, with ~20 mapped fields.

- Platform example (Kotlin + Ktor) — orchestrates the stack
- fhir-server-1 (HAPI 8.8.1) — demographics + directory store (Patient / Practitioner / Organization)
- openFHIR 2.2.3 — bidirectional FHIR ↔ openEHR mapping engine (FHIRconnect V2 YAMLS)
- EHRbase 2.31.0 — openEHR Composition store + AQL

V2 mapping: COMPOSITION-level, ~20 ACTION fields, multi-immunization Bundles, vaccination protocol, adverse reactions, performer, verification status, feeder audit.

Bidirectional mapping loaded (admin + vaccination-record); write path tested end-to-end.

The Write Flow (V2)

POST /Immunization — full Bundle in, COMPOSITION-level mapping, multi-immunization support

- 1. Client**
POSTs a CH VACD Immunization Administration Document Bundle
to platform /Immunization
- 2. Demographics fan-out**
Patient + Practitioner + Organization → fhir-server-1 (HAPI)
Platform captures the HAPI-assigned ids
- 3. EHR resolution**
Find-or-create openEHR EHR in EHRbase
Linked via EHR_STATUS.subject.external_ref (namespace=ch-vacd, id=<Patient.id>)
- 4. V2 Conversion**
Full Bundle → openFHIR /openfhir/toopenehr (COMPOSITION-level mapping)
openFHIR resolves urn:uuid refs, navigates sections, maps ~20 fields per Immunization
- 5. Enrich**
Platform adds ctx/ metadata (language, territory, composer) for EHRbase
feederAudit handled by V2 mapping — putIfAbsent fallback for Konkretisierung §13
- 6. Persist**
POST /ehr/{ehrId}/composition?format=FLAT&templateId=...
Returns 201 + compositionUid; one Composition may contain multiple ACTION entries

Demo UI — initial state

/demo loads with the first example pre-populated.

● CH VACD

Pattern A — Platform Business Logic Layer (tracer bullet)

healthz metadata CH VACD IG ↗

Example: Immunization administration — boostrix

Convert & Store ▶

1 · FHIR Immunization (input)

application/fhir+json

```
{
  "resourceType": "Bundle",
  "id": "1-1-ImmunizationAdministration",
  "meta": {
    "lastUpdated": "2021-06-01T00:00:00.374+02:00",
    "profile": ["http://fhir.ch/ig/ch-vacd/StructureDefinition/ch-vacd-document-immunization-administration"]
  },
  "identifier": {
    "system": "urn:ietf:rfc:3986",
    "value": "urn:uuid:5bca53b2-0973-4b91-961e-b1e93474c9c2"
  },
  "type": "document",
  "timestamp": "2021-06-01T00:00:00.390+02:00",
  "entry": [{
    "fullUrl":
"http://test.fhir.ch/r4/Composition/2-1-ImmunizationAdministrationComposition",
    "resource": {
      "resourceType": "Composition",
      "id": "2-1-ImmunizationAdministrationComposition",
      "meta": {
```

2 · openEHR FLAT Composition (after openFHIR)

intermediate

FLAT JSON will appear here.

3 · EHRbase confirmation

AQL roundtrip

Composition UID, EHR ID, AQL result.

How this works

1. Client POSTs a CH VACD Immunization Administration Document Bundle to `POST /Immunization`.
2. The platform fans Patient/Practitioner/Organization to `fhir-server-1` (HAPI). The FHIR server is the demographics + directory store.
3. The platform ensures an openEHR EHR exists in `EHRbase`, linked to the Patient via `EHR_STATUS.subject.external_ref`.
4. The platform sends the full Bundle to `openFHIR /openfhir/toopenehr` — the V2 FHIRconnect mapping converts it to a FLAT openEHR Composition (COMPOSITION-level mapping with ~20 fields).
5. The platform enriches FLAT with `ctx/` metadata (language, territory, composer) that EHRbase requires.
6. The platform persists the Composition to EHRbase and returns `201 Created` with `compositionUid`.

Hackathon harness — services: HAPI 8.8.1 · openFHIR 2.2.3 · EHRbase 2.31.0

screenshot: 00-initial.png

Boostrix (dTpa) — single immunization

V1 example: Swissmedic 637, deltoid IM, 1 Immunization in Bundle.

● CH VACD Pattern A — Platform Business Logic Layer (tracer bullet)

healthz metadata CH VACD IG ↗

Example: Immunization administration — boostrix

Convert & Store ▶

1 · FHIR Immunization (input)

application/fhir+json

```
{
  "resourceType": "Bundle",
  "id": "1-1-ImmunizationAdministration",
  "meta": {
    "lastUpdated": "2021-06-01T00:00:00.374+02:00",
    "profile": ["http://fhir.ch/ig/ch-vacd/StructureDefinition/ch-vacd-document-immunization-administration"]
  },
  "identifier": {
    "system": "urn:ietf:rfc:3986",
    "value": "urn:uuid:5bca53b2-0973-4b91-961e-b1e93474c9c2"
  },
  "type": "document",
  "timestamp": "2021-06-01T00:00:00.390+02:00",
  "entry": [{
    "fullUrl":
"http://test.fhir.ch/r4/Composition/2-1-ImmunizationAdministrationComposition",
    "resource": {
      "resourceType": "Composition",
      "id": "2-1-ImmunizationAdministrationComposition",
      "meta": {
```

2 · openEHR FLAT Composition (after openFHIR)

intermediate

```
{
  "ch_vacd_immunization_administration/medication
_management:0/time": "2017-09-15T07:12:56",
  "ch_vacd_immunization_administration/medication
_management:0/ism_transition/careflow_step|terminol
ogy": "local",
  "ch_vacd_immunization_administration/medication
_management:0/ism_transition/careflow_step|code":
"at0006",
  "ch_vacd_immunization_administration/medication
_management:0/ism_transition/careflow_step|value":
"Dose administration",
  "ch_vacd_immunization_administration/medication
_management:0/other_participation:0|id": "Practiti
onerRole/6-1-PractitionerRole",
  "ch_vacd_immunization_administration/medication
_management:0/other_participation:0|id_namespace":
"http://hl7.org/fhir",
  "ch_vacd_immunization_administration/medication
_management:0/other_participation:0|id_scheme": "P
ractitionerRole",
  "ch_vacd_immunization_administration/medication
_management:0/vaccine_administered|code": "637",
  "ch_vacd_immunization_administration/medication
_management:0/vaccine_administered|terminology": "h
ttp://fhir.ch/ig/ch-vacd/CodeSystem/ch-vacd-swissme
```

3 · EHRbase confirmation

AQL roundtrip

compositionUid	6f04594e-5d51-4514-858d-a3158f33b79c::local.ehrbase.org::1
ehrId	a89c937b-1f8b-4928-b824-3917b109c88b
patientId (HAPI)	f0fd8eff-9ce3-4095-99b7-b342cecf637c
practitionerIds	9d5c2cd9-9c4c-4506-a1e2-c18fcea10854
organizationIds	2a95ffe5-c60c-4eb2-af26-880c2121c075

AQL roundtrip

```
{
  "meta": {
    "type": "RESULTSET",
    "schema version": "1.0.3",
    "_created": "2026-05-26T19:30:57.00480675Z",
    "_executed_aql": "SELECT c/uid/value AS uid, c/archetype_details/template_id/value AS template FROM EHR e|ehr_id/value = 'a89c937b-1f8b-4928-b824-3917b109c88b' CONTAINS COMPOSITION c",
    "resultsize": 1,
    "q": "SELECT c/uid/value AS uid, c/archetype_details/template_id/value AS template FROM EHR e CONTAINS COMPOSITION c WHERE e/ehr_id/value = 'a89c937b-1f8b-4928-b824-3917b109c88b'",
    "columns": [
      {
        "path": "c/uid/value",
        "name": "uid",
        "path": "c/archetype_details/template_id/value",
        "name": "template"
      }
    ]
  },
  "rows": [
    {
      "uid": "a89c937b-1f8b-4928-b824-3917b109c88b",
      "template": "2-1-ImmunizationAdministrationComposition"
    }
  ]
}
```

How this works

- Client POSTs a CH VACD Immunization Administration Document Bundle to `POST /Immunization`.
- The platform fans Patient/Practitioner/Organization to `fhir-server-1` (HAPI). The FHIR server is the demographics + directory store.
- The platform ensures an openEHR EHR exists in `EHRbase`, linked to the Patient via `EHR_STATUS.subject.external_ref`.
- The platform sends the full Bundle to `openFHIR /openfhir/toopenehr` — the V2 FHIRconnect mapping converts it to a FLAT openEHR Composition (COMPOSITION-level mapping with ~20 fields).
- The platform enriches FLAT with `ctx/` metadata (language, territory, composer) that EHRbase requires.
- The platform persists the Composition to EHRbase and returns `201 Created` with `compositionUid`.

Hackathon harness — services: HAPI 8.8.1 · openFHIR 2.2.3 · EHRbase 2.31.0

screenshot: 02-01-immunization-administration-boostrix-done.png

Built 2026-05-26 — Swiss GovTech Hackathon 2026 / DigiSanté

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Comirnaty (COVID-19 mRNA) — single immunization

V1 example: Swissmedic 68225, SNOMED 840539006 (COVID-19).

● CH VACD Pattern A — Platform Business Logic Layer (tracer bullet)

healthz metadata CH VACD IG ↗

Example: Immunization administration — comirnaty ▼

Convert & Store ▶

1 · FHIR Immunization (input)

application/fhir+json

```
{
  "resourceType": "Bundle",
  "id": "1-2-ImmunizationAdministration",
  "meta": {
    "lastUpdated": "2021-06-01T00:00:00.374+02:00",
    "profile": [
      "http://fhir.ch/ig/ch-vacd/StructureDefinition/ch-vacd-document-immunization-administration"
    ]
  },
  "identifier": {
    "system": "urn:ietf:rfc:3986",
    "value": "urn:uuid:5bca53b2-0973-4b91-961e-b1e93474c9c2-derived"
  },
  "type": "document",
  "timestamp": "2024-10-21T10:30:00+02:00",
  "entry": [
    {
      "fullUrl":
"http://test.fhir.ch/r4/Composition/2-2-ImmunizationAdministrationComposition",
      "resource": {
        "resourceType": "Composition",
```

2 · openEHR FLAT Composition (after openFHIR)

intermediate

```
{
  "ch_vacd_immunization_administration/medication
_management:0/time": "2024-10-21T08:30:00",
  "ch_vacd_immunization_administration/medication
_management:0/ism_transition/careflow_step|terminol
ogy": "local",
  "ch_vacd_immunization_administration/medication
_management:0/ism_transition/careflow_step|code":
"at0006",
  "ch_vacd_immunization_administration/medication
_management:0/ism_transition/careflow_step|value":
"Dose administration",
  "ch_vacd_immunization_administration/medication
_management:0/ other_participation:0|id": "Practiti
onerRole/6-2-PractitionerRole",
  "ch_vacd_immunization_administration/medication
_management:0/ other_participation:0|id_namespace":
"http://hl7.org/fhir",
  "ch_vacd_immunization_administration/medication
_management:0/ other_participation:0|id_scheme": "P
ractitionerRole",
  "ch_vacd_immunization_administration/medication
_management:0/vaccine_administered|code": "68225",
  "ch_vacd_immunization_administration/medication
_management:0/vaccine_administered|terminology": "h
ttp://fhir.ch/ig/ch-vacd/CodeSystem/ch-vacd-swissme
```

3 · EHRbase confirmation

AQL roundtrip

compositionUid	19566356-43c0-493a-9172-ca4a0e db1bd3::local.ehrbase.org::1
ehrId	1895fa02-clb7-4afc-aafc-b6ae18 37acfc
patientId (HAPI)	ae693350-419a-4380-98da-96487f 5bdef9
practitionerIds	5ea695db-6b91-4328-a3eb-eac56a a87977
organizationIds	f0a1fd71-7875-4055-a10f-ed97c8 2397c8

AQL roundtrip

```
{
  "meta": {
    "type": "RESULTSET",
    "schema version": "1.0.3",
    "created": "2026-05-26T19:30:57.676377Z",
    "executed":
      "aql": "SELECT c/uid/value AS uid, c/archetype_details/
template_id/value AS template FROM EHR e[ehr_id/value
='1895fa02-clb7-4afc-aafc-b6ae1837acfc'] CONTAINS COMP
OSITION c",
    "results size": 1,
    "q": "SELECT c/uid/value AS uid, c/archetype_details/
template_id/value AS template FROM EHR e CONTAINS COMP
OSITION c WHERE e/ehr_id/value = '1895fa02-clb7-4afc-
aafc-b6ae1837acfc'",
    "columns": [
      {
        "path": "c/uid/value",
        "name": "uid",
        "path": "c/archet
ype_details/template_id/value",
        "name": "template"
      }
    ]
  },
  "rows": [
    {
      "uid": "19566356-43c0-493a-9172-ca4a0e-db1bd3::local.ehrbase.org::1",
      "template": "1895fa02-clb7-4afc-aafc-b6ae1837acfc"
    }
  ]
}
```

How this works

- Client POSTs a CH VACD Immunization Administration Document Bundle to `POST /Immunization`.
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- The platform sends the full Bundle to `openFHIR /openfhir/toopenehr` — the V2 FHIRconnect mapping converts it to a FLAT openEHR Composition (COMPOSITION-level mapping with ~20 fields).
- The platform enriches FLAT with `ctx/` metadata (language, territory, composer) that EHRbase requires.
- The platform persists the Composition to EHRbase and returns `201 Created` with `compositionUid`.

Hackathon harness — services: HAPI 8.8.1 · openFHIR 2.2.3 · EHRbase 2.31.0

screenshot: 04-02-immunization-administration-comirnaty-done.png

Priorix (MMR) — single immunization

V1 example: Swissmedic 615, SNOMED 14189004/36989005/36653000.

● CH VACD Pattern A — Platform Business Logic Layer (tracer bullet)

healthz metadata CH VACD IG ↗

Example: Immunization administration — priorix ▼

Convert & Store ▶

1 · FHIR Immunization (input)

application/fhir+json

```
{
  "resourceType": "Bundle",
  "id": "1-3-ImmunizationAdministration",
  "meta": {
    "lastUpdated": "2021-06-01T00:00:00.374+02:00",
    "profile": [
      "http://fhir.ch/ig/ch-vacd/StructureDefinition/ch-vacd-document-immunization-administration"
    ]
  },
  "identifier": {
    "system": "urn:ietf:rfc:3986",
    "value": "urn:uuid:5bca53b2-0973-4b91-961e-b1e93474c9c2-derived"
  },
  "type": "document",
  "timestamp": "2023-08-14T14:15:00+02:00",
  "entry": [
    {
      "fullUrl":
"http://test.fhir.ch/r4/Composition/2-3-ImmunizationAdministrationComposition",
      "resource": {
        "resourceType": "Composition",
```

2 · openEHR FLAT Composition (after openFHIR)

intermediate

```
{
  "ch_vacd_immunization_administration/medication
_management:0/time": "2023-08-14T12:15:00",
  "ch_vacd_immunization_administration/medication
_management:0/ism_transition/careflow_step|terminol
ogy": "local",
  "ch_vacd_immunization_administration/medication
_management:0/ism_transition/careflow_step|code":
"at0006",
  "ch_vacd_immunization_administration/medication
_management:0/ism_transition/careflow_step|value":
"Dose administration",
  "ch_vacd_immunization_administration/medication
_management:0/other_participation:0|id": "Practiti
onerRole/6-3-PractitionerRole",
  "ch_vacd_immunization_administration/medication
_management:0/other_participation:0|id_namespace":
"http://hl7.org/fhir",
  "ch_vacd_immunization_administration/medication
_management:0/other_participation:0|id_scheme": "P
ractitionerRole",
  "ch_vacd_immunization_administration/medication
_management:0/vaccine_administered|code": "615",
  "ch_vacd_immunization_administration/medication
_management:0/vaccine_administered|terminology": "h
ttp://fhir.ch/ig/ch-vacd/CodeSystem/ch-vacd-swissme
```

3 · EHRbase confirmation

AQL roundtrip

compositionUid	c3e027ac-6ee3-4d31-8313-a15579b7bc8a::local.ehrbase.org::1
ehrId	e815c28e-309a-4977-ae6c-77bf74eb9226
patientId (HAPI)	260217e4-648b-440b-a432-d92170a284b9
practitionerIds	bdda7a9f-8497-4e09-810d-87316a6dfalac
organizationIds	01cb99a4-1827-485d-b55e-a2d8e97e2fd1

AQL roundtrip

```
{
  "meta": {
    "type": "RESULTSET",
    "schema version": "1.0.3",
    "_created": "2026-05-26T19:30:58.267355626Z",
    "_executed_aql": "SELECT c/uid/value AS uid, c/archetype_details/template_id/value AS template FROM EHR e[ehr_id/value='e815c28e-309a-4977-ae6c-77bf74eb9226'] CONTAINS COMPOSITION c, \"resultsize\":1,\"q\":\"SELECT c/uid/value AS uid, c/archetype_details/template_id/value AS template FROM EHR e CONTAINS COMPOSITION c WHERE e/ehr_id/value = 'e815c28e-309a-4977-ae6c-77bf74eb9226'\", \"columns\": [{\"path\":\"c/uid/value\", \"name\":\"uid\"}, {\"path\":\"c/archetype_details/template_id/value\", \"name\":\"templat
```

How this works

- Client POSTs a CH VACD Immunization Administration Document Bundle to `POST /Immunization`.
- The platform fans Patient/Practitioner/Organization to `fhir-server-1` (HAPI). The FHIR server is the demographics + directory store.
- The platform ensures an openEHR EHR exists in `EHRbase`, linked to the Patient via `EHR_STATUS.subject.external_ref`.
- The platform sends the full Bundle to `openFHIR /openfhir/toopenehr` — the V2 FHIRconnect mapping converts it to a FLAT openEHR Composition (COMPOSITION-level mapping with ~20 fields).
- The platform enriches FLAT with `ctx/` metadata (language, territory, composer) that EHRbase requires.
- The platform persists the Composition to EHRbase and returns `201 Created` with `compositionUid`.

Hackathon harness — services: HAPI 8.8.1 · openFHIR 2.2.3 · EHRbase 2.31.0

screenshot: 06-03-immunization-administration-priorix-done.png

Built 2026-05-26 — Swiss GovTech Hackathon 2026 / DigiSanté

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V2: 2-dose Comirnaty series

Multi-immunization Bundle: 2 Comirnaty doses → 1 Composition, 2 ACTION entries.

● CH VACD Pattern A — Platform Business Logic Layer (tracer bullet)

healthz metadata CH VACD IG ↗

Example: Immunization administration — v2 2dose comirnaty ▾

Convert & Store ▶

1 · FHIR Immunization (input)

application/fhir+json

```
{
  "resourceType": "Bundle",
  "id": "20f16d0b-fdc9-423f-a3b8-38e9f45247f0",
  "meta": {
    "profile": [
      "http://fhir.ch/ig/ch-vacd/StructureDefinition/ch-vacd-document-immunization-administration"
    ]
  },
  "language": "en-US",
  "identifier": {
    "system": "urn:ietf:rfc:3986",
    "value": "urn:uuid:20f16d0b-fdc9-423f-a3b8-38e9f45247f0"
  },
  "type": "document",
  "timestamp": "2026-05-23T21:21:17.397+02:00",
  "entry": [
    {
      "fullUrl": "urn:uuid:11ddeaab-1cd5-40b9-a973-82b0bf21612c",
      "resource": {
        "resourceType": "Composition",
        "meta": {
          "profile": [
            "http://fhir.ch/ig/ch-
```

2 · openEHR FLAT Composition (after openFHIR)

intermediate

```
{
  "ch_vacd_immunization_administration/medication_management:0/time": "2021-05-22T00:00:00",
  "ch_vacd_immunization_administration/medication_management:0/ism_transition/careflow_step|terminology": "local",
  "ch_vacd_immunization_administration/medication_management:0/ism_transition/careflow_step|code": "at0006",
  "ch_vacd_immunization_administration/medication_management:0/ism_transition/careflow_step|value": "Dose administration",
  "ch_vacd_immunization_administration/medication_management:0/other_participation:0|id": "urn:uuid:80945811-f4de-454b-b868-94b577affc34",
  "ch_vacd_immunization_administration/medication_management:0/other_participation:0|id_namespace": "http://hl7.org/fhir",
  "ch_vacd_immunization_administration/medication_management:0/vaccine_administered|code": "68225",
  "ch_vacd_immunization_administration/medication_management:0/vaccine_administered|terminology": "http://fhir.ch/ig/ch-vacd/CodeSystem/ch-vacd-swissmedic-cs",
  "ch_vacd_immunization_administration/medication_management:0/vaccine_administered|value": "Comirna
```

3 · EHRbase confirmation

AQL roundtrip

compositionUid	be781748-a375-400c-8832-8574be4e57f3::local.ehrbase.org::1
ehrId	89abf891-0f98-4d18-bd36-70b61d05c90a
patientId (HAPI)	08febdb6-3d4f-4e81-828f-9ac7b1b4938d
practitionerIds	1307dac1-78df-43d5-bcb0-4be5896010fd, b689f8dd-dc35-47db-87ea-107ac08d01a6, 48171d49-222c-42a5-a563-c0d1988783c9
organizationIds	f9581ebd-2e65-453b-b8f7-0baa2a16781c, e935ab61-166e-47f9-82dd-62429bc9adff, 3425a87d-bf83-46a1-8479-7f172dd02cbe

AQL roundtrip

```
{
  "meta": {
    "type": "RESULTSET",
    "schema_version": "1.0.3",
    "created": "2026-05-26T19:30:59.152064793Z",
    "executed_aql": "SELECT c/uid/value AS uid, c/archetype_details/template_id/value AS template FROM EHR e[ehr_id/value='89abf891-0f98-4d18-bd36-70b61d05c90a'] CONTAINS COMPOSITION c",
    "resultsize": 1,
    "q": "SELECT c/uid/value AS uid, c/archetype_details/template_id/value AS template FROM EHR e CONTAINS COMPOSITION c WHERE e/ehr_id/value = '89abf891-0f98-4d18-bd36-70b61d05c90a'",
    "columns": [
      {
        "path": "c/uid/value",
        "name": "uid",
      },
      {
        "path": "c/archetype_details/template_id/value",
        "name": "templat
```

How this works

- Client POSTs a CH VACD Immunization Administration Document Bundle to `POST /Immunization`.
- The platform fans Patient/Practitioner/Organization to `fhir-server-1` (HAPI). The FHIR server is the demographics + directory store.
- The platform ensures an openEHR EHR exists in `EHRbase`, linked to the Patient via `EHR_STATUS.subject.external_ref`.
- The platform sends the full Bundle to `openFHIR /openfhir/toopenehr` — the V2 FHIRconnect mapping converts it to a FLAT openEHR Composition (COMPOSITION-level mapping with ~20 fields).
- The platform enriches FLAT with `ctx/` metadata (language, territory, composer) that EHRbase requires.
- The platform persists the Composition to EHRbase and returns `201 Created` with `compositionUid`.

Hackathon harness — services: HAPI 8.8.1 · openFHIR 2.2.3 · EHRbase 2.31.0

screenshot: 08-04-immunization-administration-v2-2dose-comirnaty-done.png

V2: 3-dose Comirnaty series

Multi-immunization Bundle: 3 Comirnaty 30µg doses → 1 Composition, 3 ACTION entries.

● CH VACD Pattern A — Platform Business Logic Layer (tracer bullet)

healthz metadata CH VACD IG ↗

Example: Immunization administration — v2 3dose comirnaty ▾

Convert & Store ▶

1 · FHIR Immunization (input)

application/fhir+json

```
{
  "resourceType": "Bundle",
  "id": "cf36b4ea-feac-47d6-ab5c-99f35a88733c",
  "meta": {
    "profile": [
      "http://fhir.ch/ig/ch-vacd/StructureDefinition/ch-vacd-document-immunization-administration"
    ]
  },
  "language": "en-US",
  "identifier": {
    "system": "urn:ietf:rfc:3986",
    "value": "urn:uuid:cf36b4ea-feac-47d6-ab5c-99f35a88733c"
  },
  "type": "document",
  "timestamp": "2025-09-18T13:37:33.094+02:00",
  "entry": [
    {
      "fullUrl": "urn:uuid:a515fdc9-f019-4376-9ba8-5d172dc3be81",
      "resource": {
        "resourceType": "Composition",
        "meta": {
          "profile": [
            "http://fhir.ch/ig/ch-vacd/StructureDefinition/ch-vacd-composition-immunization-administration"
          ]
        }
      }
    }
  ]
}
```

2 · openEHR FLAT Composition (after openFHIR)

intermediate

```
{
  "ch_vacd_immunization_administration/medication_management:0/time": "2021-04-17T00:00:00",
  "ch_vacd_immunization_administration/medication_management:0/ism_transition/careflow_step|terminology": "local",
  "ch_vacd_immunization_administration/medication_management:0/ism_transition/careflow_step|code": "at0006",
  "ch_vacd_immunization_administration/medication_management:0/ism_transition/careflow_step|value": "Dose administration",
  "ch_vacd_immunization_administration/medication_management:0/other_participation:0|id": "urn:uuid:637385aa-88e3-4d90-aed6-cd01f1306a02",
  "ch_vacd_immunization_administration/medication_management:0/other_participation:0|id_namespace": "http://hl7.org/fhir",
  "ch_vacd_immunization_administration/medication_management:0/vaccine_administered|code": "68710-01",
  "ch_vacd_immunization_administration/medication_management:0/vaccine_administered|terminology": "http://fhir.ch/ig/ch-vacd/CodeSystem/ch-vacd-swissmedic-cs",
  "ch_vacd_immunization_administration/medication_management:0/vaccine_administered|value": "68710-01"
}
```

3 · EHRbase confirmation

AQL roundtrip

compositionUId	bf9fc6a9-5875-4fe9-b87c-b68fa3ae3c6d::local.ehrbase.org::1
ehrId	0498aa6d-9037-4049-9fd3-2c41663fed9f
patientId (HAPI)	7eb221f3-724e-4e7c-8846-59fd5d1bf6a3
practitionerIds	0047c989-50cc-4127-af7e-19d06eb476a9, 7bc3f022-3bb6-43cb-91a8-4ee2319f31f4, 4f60fcec-8fcc-43aa-9523-7bd9bdb0325b, 436e0340-4101-456d-aa33-59da411539f5
organizationIds	83a78ea6-3463-40ee-9c5b-feb60d611d07, 0f13e3fc-d2b0-42f5-bdfa-f7be9aef3eed, 520b801a-b175-4393-ad2c-112256860f82, bda3ea-fa-778b-4b81-89be-a01965f66345

AQL roundtrip

```
{
  "meta": {
    "type": "RESULTSET",
    "schema_version": "1.0.3",
    "created": "2026-05-26T19:31:00.189224418Z",
    "executed_aql": "SELECT c/uid/value AS uid, c/archetype_details/template_id/value AS template FROM EHR e[ehr_id/value='0498aa6d-9037-4049-9fd3-2c41663fed9f'] CONTAINS COMPOSITION c, \"resultsize\":1}, \"q\": \"SELECT c/uid/value AS uid, c/archetype_details/template_id/value AS template FROM EHR e CONTAINS COMPOSITION c WHERE e/ehr_id/value = '0498aa6d-9037-4049-9fd3-2c41663fed9f', \"columns\": [{\"path\": \"c/uid/value\", \"name\": \"uid\"}, {\"path\": \"c/archetype_details/template_id/value\", \"name\": \"templat\"}]"
  }
}
```

How this works

- Client POSTs a CH VACD Immunization Administration Document Bundle to `POST /Immunization`.
- The platform fans Patient/Practitioner/Organization to `fhir-server-1` (HAPI). The FHIR server is the demographics + directory store.
- The platform ensures an openEHR EHR exists in `EHRbase`, linked to the Patient via `EHR_STATUS.subject.external_ref`.
- The platform sends the full Bundle to `openFHIR /openfhir/toopenehr` — the V2 FHIRconnect mapping converts it to a FLAT openEHR Composition (COMPOSITION-level mapping with ~20 fields).
- The platform enriches FLAT with `ctx/` metadata (language, territory, composer) that EHRbase requires.
- The platform persists the Composition to EHRbase and returns `201 Created` with `compositionUId`.

Hackathon harness — services: HAPI 8.8.1 · openFHIR 2.2.3 · EHRbase 2.31.0

screenshot: 10-05-immunization-administration-v2-3dose-comirnaty-done.png

Built 2026-05-26 — Swiss GovTech Hackathon 2026 / DigiSanté

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V2: Mixed vaccines (Boostrix Polio + FSME)

Multi-immunization Bundle: 2× Boostrix Polio + 1× FSME-Immun Junior → 3 ACTIONS, different vaccine codes.

● CH VACD Pattern A — Platform Business Logic Layer (tracer bullet)

healthz metadata CH VACD IG ↗

Example: Immunization administration — v2 3dose mixed ▾

Convert & Store ▶

1 · FHIR Immunization (input)

application/fhir+json

```
{
  "resourceType": "Bundle",
  "id": "3c12e373-0c33-4fal-9067-ca7de785477d",
  "meta": {
    "profile": [
      "http://fhir.ch/ig/ch-vacd/StructureDefinition/ch-vacd-document-immunization-administration"
    ]
  },
  "language": "en-US",
  "identifier": {
    "system": "urn:ietf:rfc:3986",
    "value": "urn:uuid:3c12e373-0c33-4fal-9067-ca7de785477d"
  },
  "type": "document",
  "timestamp": "2026-05-23T21:31:58.228+02:00",
  "entry": [
    {
      "fullUrl": "urn:uuid:fb68fc23-a151-4ea0-a575-a9a59a8b10eb",
      "resource": {
        "resourceType": "Composition",
        "meta": {
          "profile": [
            "http://fhir.ch/ig/ch-
```

2 · openEHR FLAT Composition (after openFHIR)

intermediate

```
{
  "ch_vacd_immunization_administration/medication_management:0/time": "2009-04-22T00:00:00",
  "ch_vacd_immunization_administration/medication_management:0/ism_transition/careflow_step|terminology": "local",
  "ch_vacd_immunization_administration/medication_management:0/ism_transition/careflow_step|code": "at0006",
  "ch_vacd_immunization_administration/medication_management:0/ism_transition/careflow_step|value": "Dose administration",
  "ch_vacd_immunization_administration/medication_management:0/other_participation:0|id": "urn:uuid:0faa55b6-0cc2-44ea-a74e-cc8b2b4ae341",
  "ch_vacd_immunization_administration/medication_management:0/other_participation:0|id_namespace": "http://hl7.org/fhir",
  "ch_vacd_immunization_administration/medication_management:0/vaccine_administered|code": "681",
  "ch_vacd_immunization_administration/medication_management:0/vaccine_administered|terminology": "http://fhir.ch/ig/ch-vacd/CodeSystem/ch-vacd-swissmedic-cs",
  "ch_vacd_immunization_administration/medication_management:0/vaccine_administered|value": "Boostrix
```

3 · EHRbase confirmation

AQL roundtrip

compositionUid	c8439da3-ad6f-42a0-8112-040a89b6eb4e::local.ehrbase.org::1
ehrId	d204ade0-d570-4c91-b627-c085110100e3
patientId (HAPI)	0c96531a-ea14-41a4-8cca-1ccd5e0badb5
practitionerIds	76791bab-8a60-4089-94c5-a6fa724bd556, 09967fd3-a0c9-45e0-a2d1-90c8b283050a, 79b20998-a185-462a-97ee-17e75f31db92
organizationIds	2eb542d0-58f3-4b5b-9f54-48037cb41969, 8f94962b-3307-4c2d-b163-a5441b0f934c, 57d67692-b536-4261-ae6d-fdae91cc419a

AQL roundtrip

```
{
  "meta": {
    "type": "RESULTSET",
    "schema version": "1.0.3",
    "created": "2026-05-26T19:31:01.240870294Z",
    "executed_aql": "SELECT c/uid/value AS uid, c/archetype_details/template_id/value AS template FROM EHR e[ehr_id/value='d204ade0-d570-4c91-b627-c085110100e3'] CONTAINS COMPOSITION c, \"resultsize\":1,\"q\":\"SELECT c/uid/value AS uid, c/archetype_details/template_id/value AS template FROM EHR e CONTAINS COMPOSITION c WHERE e/ehr_id/value = 'd204ade0-d570-4c91-b627-c085110100e3', \"columns\": [{\"path\": \"c/uid/value\", \"name\": \"uid\"}, {\"path\": \"c/archetype_details/template_id/value\", \"name\": \"template\"}]"
  }
}
```

How this works

- Client POSTs a CH VACD Immunization Administration Document Bundle to `POST /Immunization`.
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- The platform enriches FLAT with `ctx/` metadata (language, territory, composer) that EHRbase requires.
- The platform persists the Composition to EHRbase and returns `201 Created` with `compositionUid`.

Hackathon harness — services: HAPI 8.8.1 · openFHIR 2.2.3 · EHRbase 2.31.0

screenshot: 12-06-immunization-administration-v2-3dose-mixed-done.png

What's Proven

All validated end-to-end against the live stack with integration tests + AQL queries.

Proven (all green)

- ✓ V2 FHIRconnect mapping loaded as bidirectional unit (2 OPTs, 2 contexts, 4 models)
- ✓ openFHIR 2.2.3 COMPOSITION-level mapping: full Bundle in, ~20 fields per Immunization
- ✓ Multi-immunization Bundles (2-dose, 3-dose series) → single Composition, multiple ACTIONS
- ✓ Mixed vaccines (Boostrix Polio + FSME-Immun) in one Composition — different codes per ACTION
- ✓ AQL roundtrip: vaccineCode, time, ISM transition, doseNumber verified via queries
- ✓ feeder_audit: V2 mapping embeds original FHIR at COMPOSITION + ACTION level
- ✓ Performer participation: openFHIR maps actor ref, platform fills required RM fields
- ✓ V1 single-immunization Bundles still work (backward compatible)
- ✓ Demographics (Patient/Practitioner/Organization) landed in HAPI from document Bundles
- ✓ 12/12 tests pass (8 unit + 4 integration)

Next steps

- Read path — vaccination-record template + tofhir mapping (loaded, not yet tested)
- Profile validation — HAPI FhirValidator + CH VACD StructureDefinitions
- Terminology lookups — live tx.fhir.ch/r4 for code system validation
- Pattern B — inbound/outbound connectors for producer/consumer split